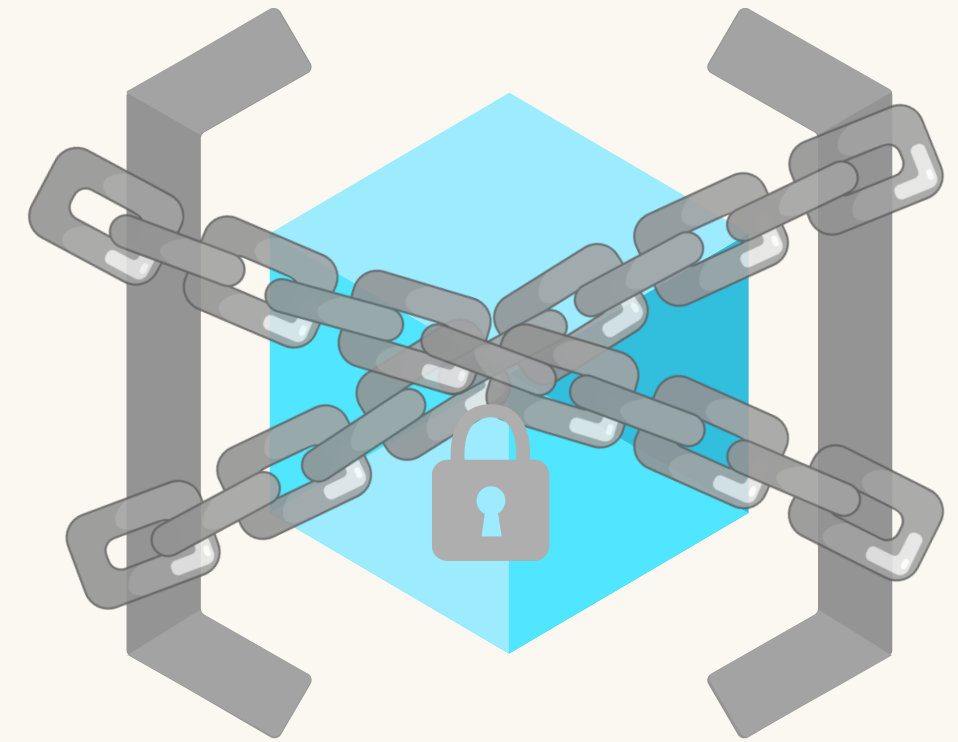




Avoid this common
mistake in
Azure Kubernetes
Service (AKS)
with the

NRGLockdown
feature



Today, we'll learn and cover:



1.

An overview of
resource
groups in AKS

2.

An overview of
the unsupported
node resource
group actions

3.

How to perform
a common
mistake in AKS

4.

Frequent needs
and the right
way to achieve
them

5.

What
NRGLockdown
is how it works

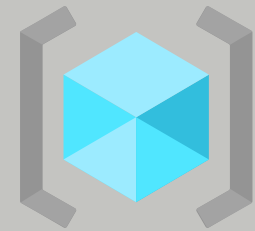
6.

How to
implement the
NRGLockdown
feature, test
and explore it

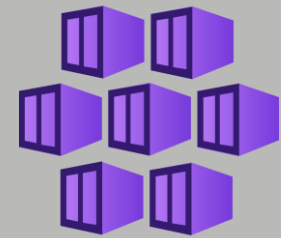
7.

What can be
done if the
mistake was
already
performed

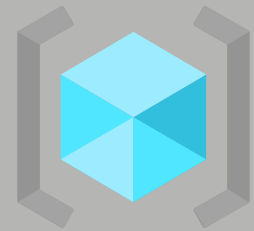
1. An overview of resource groups in AKS



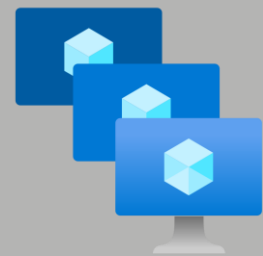
AKS Resource Group



AKS Cluster



Infrastructure / Node Resource Group



VMSS



VNET



Load Balancer



NSG



Route Table



Public IP



Managed
Identity

2. An overview of the unsupported node resource group actions

The following **direct actions** are **unsupported** for the resources part of the node/infrastructure resource group:

Deletions

Modifications / configurations

Considerations

!

Direct actions involve performing operations on resources, such as using IaaS APIs in the Azure Portal or resource-specific commands like "az vmss" and "az network lb"

!

Direct actions **won't persist** through PUT operations like upgrade, reconcile, etc.

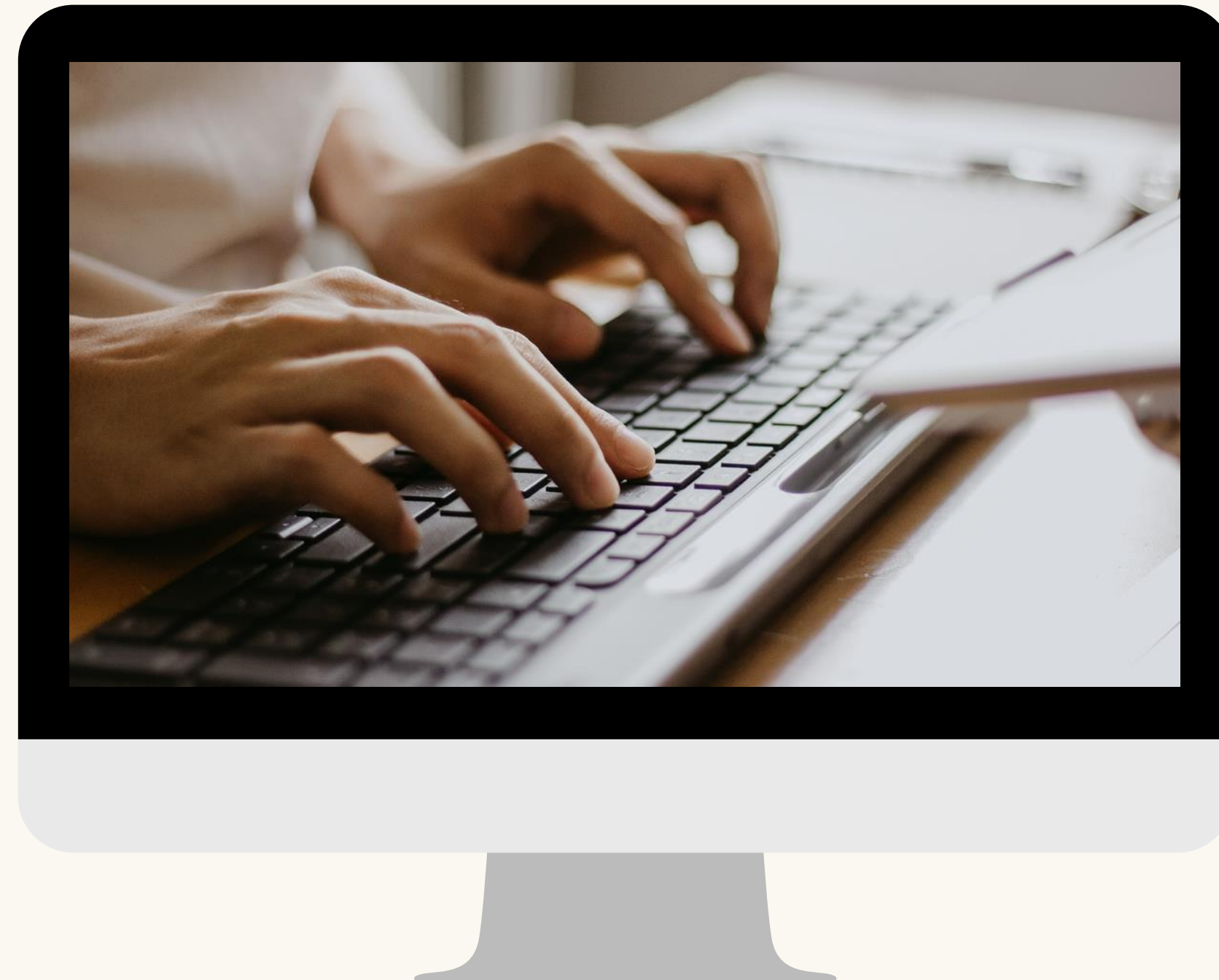
!

Any modification or deletion (if possible) should be performed **via Kubernetes or the AKS API**, for example, "kubectl" or "az aks" commands

!

Most of the resources (for example, the VNET, managed identity, etc.) **should not be deleted** in any way

3. How to perform a common mistake in AKS



4. Frequent needs and the right way to achieve them

THE NEED

HOW TO ACHIEVE IT

Have a new VMSS / delete an existing VMSS

Implement VM autoscaling

Configure the Load Balancer or the Public IP

Deploy an Internal Load Balancer

Delete the Public Load Balancer and IP

Create a new node pool / delete an existing node pool

Enable the Cluster Autoscaler

Modify the Kubernetes service or find the proper "az aks" command (depending on the goal)

Create a Kubernetes service with the internal load balancer annotation set to "true"

Update the outbound type

5. What NRGLockdown is how it works

NRGLockdown sets the restriction level for the node / infrastructure resource group.

Restriction levels:

Unrestricted

- Allows the direct modification or deletion of the resources
- Flexible but **risky**

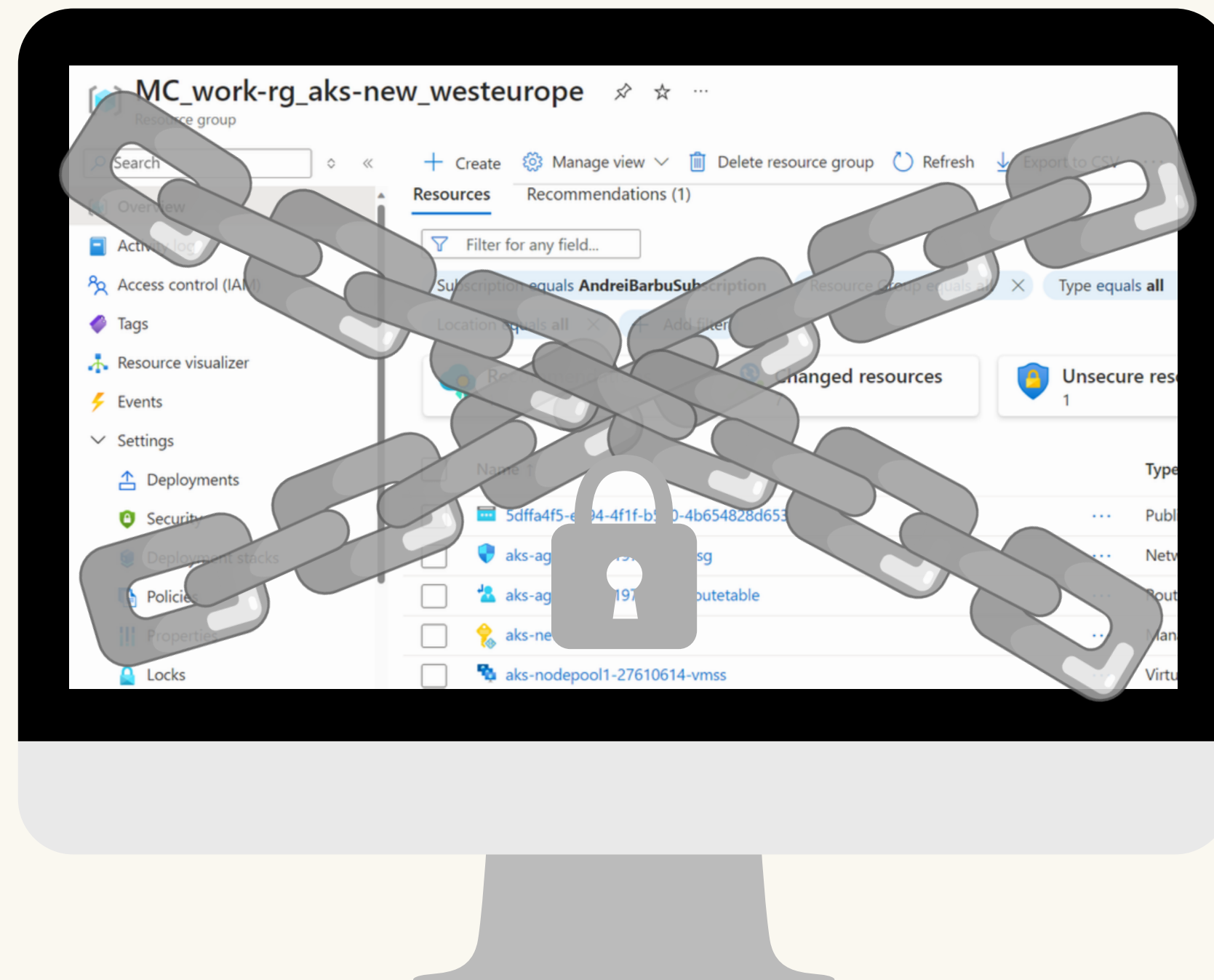
ReadOnly

- Prevents direct resource modification or deletion by configuring a **deny assignment**
- Restrictive but **safe**

Fully managed resource group



6. How to implement the NRGLockdown feature, test and explore it



7. What can be done if the mistake was already performed



Find the **right way** to perform the needed action

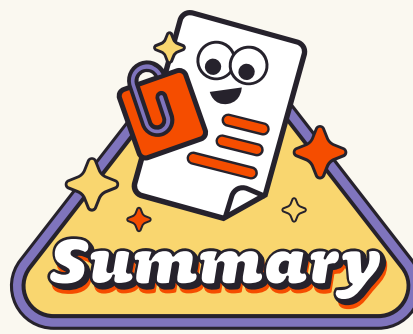


Consider **reconciling** the AKS cluster



Get in touch with **Microsoft Support**

Today, we've learned and covered:



The AKS-related
resource groups



The unsupported
node resource
group actions
and performed
one



The right way
to perform
changes and a
few examples



What
NRGLockdown is,
how to
implement it and
how it works



What can be
done if the
mistake was
already
performed

